

KITEWRIGHT CORE · REQUIREMENTS

Kitewright Core requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-core		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	TRL 3 status of every requirement from KWC-CAL-001; rated payload class and bus voltage window stated (KWC-DDR-001); targets unchanged
0.1	2026-09-30	Amish Chadha	Initial scaffold

Kitewright Core requirements

The requirements below are unchanged in their targets from version 0.1. Each now carries its TRL 3 status from the sizing calculations, KWC-CAL-001, on the constructable design of KWC-DDR-002. Ten are met on paper or by design; one, R9 core mass, is not met and is set out for Amish's decision in docs/REVIEW.md .

Safety: The core carries lithium pack power of up to 60 V and 200 A, and holds payloads that would fall if released in flight. Requirements R3 and R5 and the build plan's safety stops cover these; none of the figures here is a safety approval.

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	TRL 3 status (KWC-CAL-001)
R1	Payload electrical interface	DS-014 40-pin connector with all required signals and battery power present and documented	Continuity and protocol test with a reference payload on the bench	Met on paper; licence and pinout to confirm from the standard
R2	Payload swap	Swap in 60 s or less without tools, with locking pin engaged and visible (target)	Timed field trials with gloved users	Met on paper: about 52 s with gloves (estimate)
R3	Mount strength	Holds 3 times the rated payload mass statically and 1.5 times under a 2 g drop (target)	CalRig proof-load and drop test	Met on paper: factor 10 or more on every part at 147 N
R4	Payload power	At least 100 W continuous to the payload at bus voltage (target)	Load bank test at -20 and +45 °C	Met on paper: at most 5.6 A; DS-014 pin rating to confirm
R5	Operating temperature	Avionics boot and fly from -20 to +45 °C (target)	Climatic chamber soak and bench run	Met on paper: hottest part about 75 °C, cold boot within rating
R6	Altitude envelope	Stable control and failsafes at air density equivalent to 5,000 m (target)	AltiRig tests, then field flights at a high site	Met on paper for the core; flight check needs a frame
R7	Performance model	Published hover time model predicts measured hover time within 15% at each altitude band (target)	Compare model to AltiRig and flight logs	Model published; accuracy shown only at TRL 4
R8	Common core	Same core flies Lift and Range with parameter changes only	Build both frames and fly from one core set	Met by design
R9	Core mass	Avionics, radios and mount 1.0 kg or less, excluding batteries (target)	Weigh the assembled core	Not met: 1.27 kg estimated (Decisions for Amish)
R10	Upstream firmware	Runs a released PX4 version with no source changes, parameters and modules only	Build check against the PX4 release tag	Met by design
R11	Prototype cost	USD 5,000 or less including ground station and spares	Bill of materials and receipts	USD 3,940.60, under the value-engineering target by USD 1,059.40

■ Definitions fixed at TRL 3

- **Rated payload:** 5 kg on the mount, one payload class for version 1, so R3's load cases are 147 N vertical (KWC-DDR-001, D4).
- **Bus voltage:** 18 to 60 V at the core's pack inputs, from 6S Li-ion to 16S LiFePO4 (KWC-DDR-001, D3).
- **Core mass (R9):** the airborne core as built, excluding batteries, the payload shoe (each payload carries its own) and the optional companion computer (carried outside the core, KWC-DDR-001, D8).
- **Prototype cost (R11):** treated as a value-engineering target, not a limit (STANDARDS section 18).

■ Requirements at risk

- R9 is not met on paper; options and a recommendation are in docs/REVIEW.md , TRL 3 section, Decisions for Amish.
- R2 has a small margin (about 52 s); a gloved timed trial at TRL 4 decides it.
- R1 and R4 depend on the DS-014 licence, pinout and pin rating, to be confirmed from the published standard (register, To confirm).

■ Assumptions

- The 47% hover time at 5,000 m and -20 °C is a first estimate, not a measurement; KWC-CAL-001 traces it to unheated packs and gives 70 % with ColdCell packs.
- Pixhawk-standard flight controllers and DS-014 connectors can be bought in small quantities.
- PX4 supports the needed multicopter and quadplane modes without source changes.
- ColdCell can keep packs above their minimum discharge temperature for a full flight.